

117. Let X and Y represent prices (in ₹) of a commodity in Kolkata and Mumbai respectively. It is given that $\bar{X} = 65$, $\bar{Y} = 67$, $\sigma_X = 2.5$, $\sigma_Y = 3.5$ and $r(X, Y) = 0.8$. What is the equation of regression of Y on X ?

(a) $Y = 0.175X - 5$

(b) $Y = 1.12X - 5.8$

(c) $Y = 1.12X - 5$

(d) $Y = 0.17X + 5.8$

118. Consider a random variable X which follows Binomial distribution with parameters $n = 10$ and $p = \frac{1}{5}$. Then $Y = 10 - X$ follows Binomial distribution with parameters n and p respectively given by

(a) $5, \frac{1}{5}$

(b) $5, \frac{2}{5}$

(c) $10, \frac{3}{5}$

(d) $10, \frac{4}{5}$

119. If A and B are two events such that $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cap B) = 0.4$, then consider the following statements:

1. $P(\bar{A} \cup B) = 0.9$.

2. $P(\bar{B} | \bar{A}) = 0.6$.

Which of the above statements is/are correct?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

120. Three cooks X , Y and Z bake a special kind of cake, and with respective probabilities 0.02, 0.03 and 0.05, it fails to rise. In the restaurant where they work, X bakes 50%, Y bakes 30% and Z bakes 20% of cakes. What is the proportion of failures caused by X ?

(a) $\frac{9}{29}$

(b) $\frac{10}{29}$

(c) $\frac{19}{29}$

(d) $\frac{28}{29}$

कच्चे काम के लिए जगह
SPACE FOR ROUGH WORK